# CSE 150A-250A AI: Probabilistic Models

## Lecture 7

Fall 2026

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

# Agenda

Review

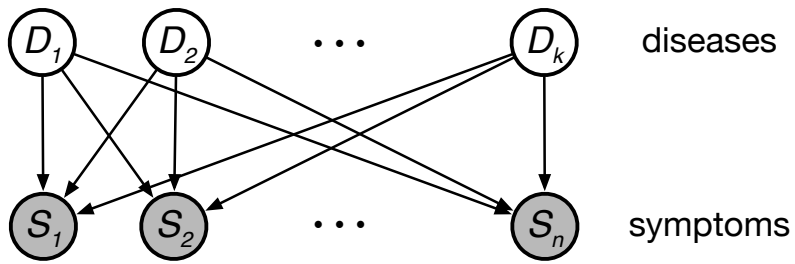
Markov chain Monte Carlo

# Review

# Approximate inference

- **Problem (for loopy BNs)**

  Given a set $E$ of evidence nodes, and a set $Q$ of query nodes, how to estimate the posterior distribution $P(Q|E)$?




- **Stochastic sampling methods**

  LAST CLASS

  1. Rejection sampling — **slow**
  2. Likelihood weighting — **faster**

  TODAY

  3. Markov chain Monte Carlo (MCMC) — **fastest**

# Likelihood weighting

- **Make $N$ forward passes through the BN:**

  Sample non-evidence nodes based on values of parents.
  Fix evidence nodes to desired values.

- **For single query and evidence nodes:**

$$P(Q=q|E=e) \approx \frac{\sum_{i=1}^{N} I(q,q_i)\overbrace{P(E=e|\mathrm{pa}_i(E))}^{\text{likelihood weight}}}{\sum_{i=1}^{N} P(E=e|\mathrm{pa}_i(E))}$$

- **For multiple query and evidence nodes:**

$$P(Q=q, Q'=q'|E=e, E'=e')$$
$$\approx \frac{\sum_{i=1}^{N} I(q,q_i)\, I(q',q'_i)\, P(E=e|\mathrm{pa}_i(E))\, P(E'=e'|\mathrm{pa}_i(E'))}{\sum_{i=1}^{N} P(E=e|\mathrm{pa}_i(E))\, P(E'=e'|\mathrm{pa}_i(E'))}$$

# Example for likelihood weighting sampling

**Problem:** Estimate $P(a_0|c_1, d_1)$

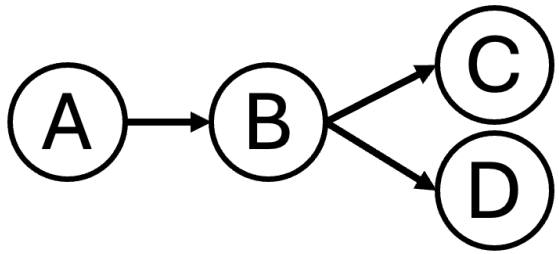
**Samples:**

$a_0, b_1, c_1, d_1$

$a_1, b_0, c_1, d_1$

$a_0, b_1, c_1, d_1$

**Q.** Estimate of $P(a_0|c_1, d_1)$ using likelihood weighting?




| A | P(A) |
|---|---|
| $a_0$ | 1/5 |
| $a_1$ | 4/5 |

| A | B | P(B\|A) |
|---|---|---|
| $a_0$ | $b_0$ | 1/4 |
| $a_0$ | $b_1$ | 3/4 |
| $a_1$ | $b_0$ | 1/3 |
| $a_1$ | $b_1$ | 2/3 |

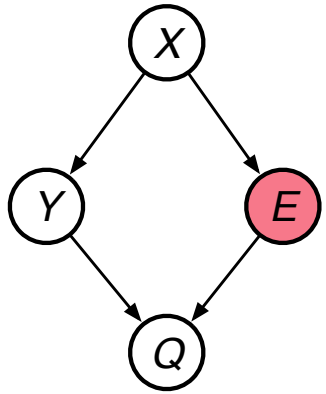
| B | C | P(C\|B) |
|---|---|---|
| $b_0$ | $c_0$ | 1/5 |
| $b_0$ | $c_1$ | 4/5 |
| $b_1$ | $c_0$ | 3/5 |
| $b_1$ | $c_1$ | 2/5 |

| B | D | P(D\|B) |
|---|---|---|
| $b_0$ | $d_0$ | 3/4 |
| $b_0$ | $d_1$ | 1/4 |
| $b_1$ | $d_0$ | 1/3 |
| $b_1$ | $d_1$ | 2/3 |

# Properties of likelihood weighting

- It converges in the limit:

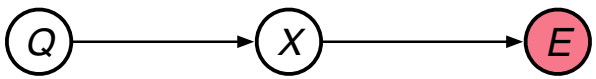
$$\lim_{N\to\infty} \frac{\sum_{i=1}^{N} l(q, q_i)\, P(E=e|X=x_i)}{\sum_{i=1}^{N} P(E=e|X=x_i)} = P(Q=q|E=e)$$




- It's more efficient than rejection sampling:

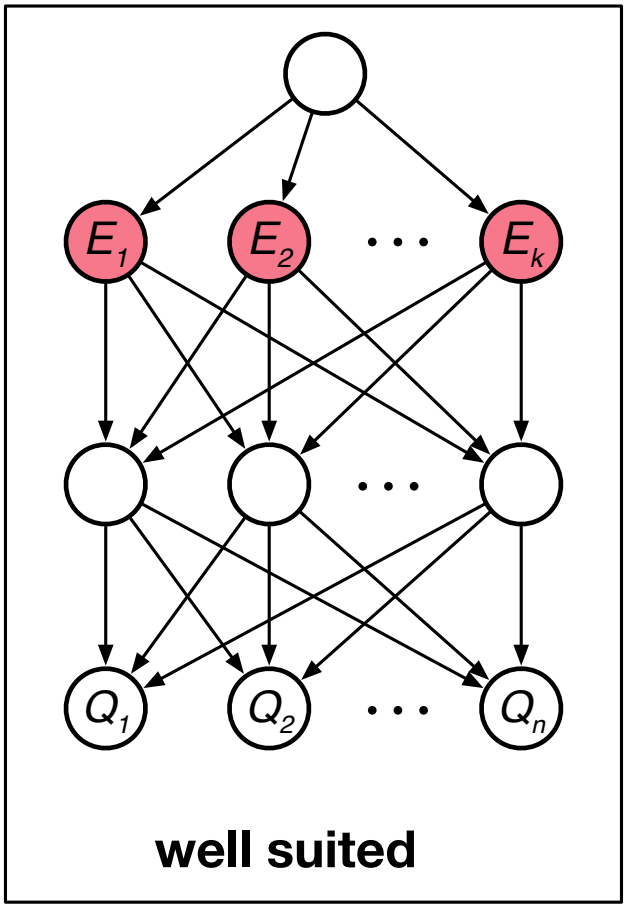
  No samples need to discarded.
  Descendants of evidence nodes are conditioned on evidence.

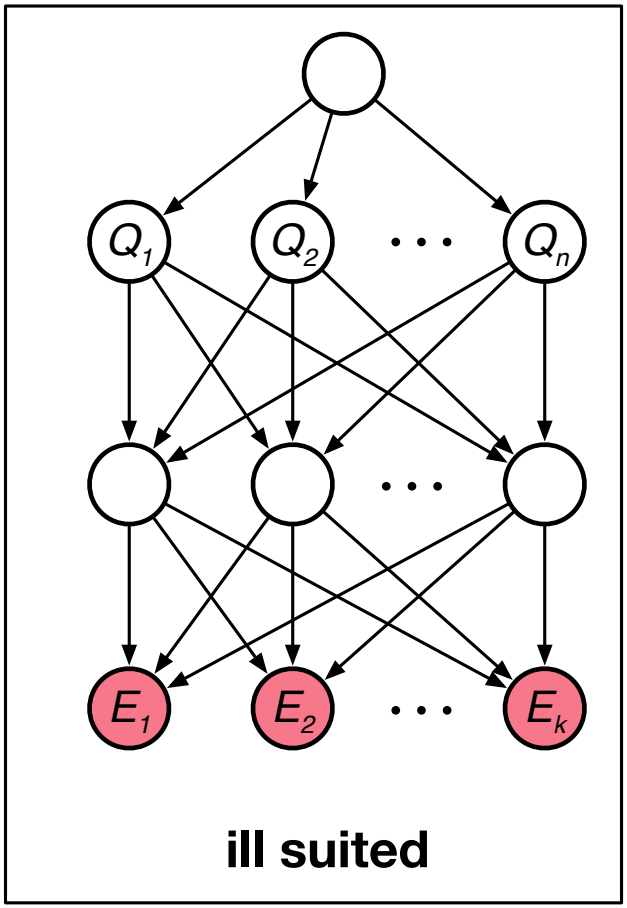
- But it can still be very slow:




The worst case for likelihood weighting is when rare evidence is descended from query nodes.

# Best and worst cases for likelihood weighting



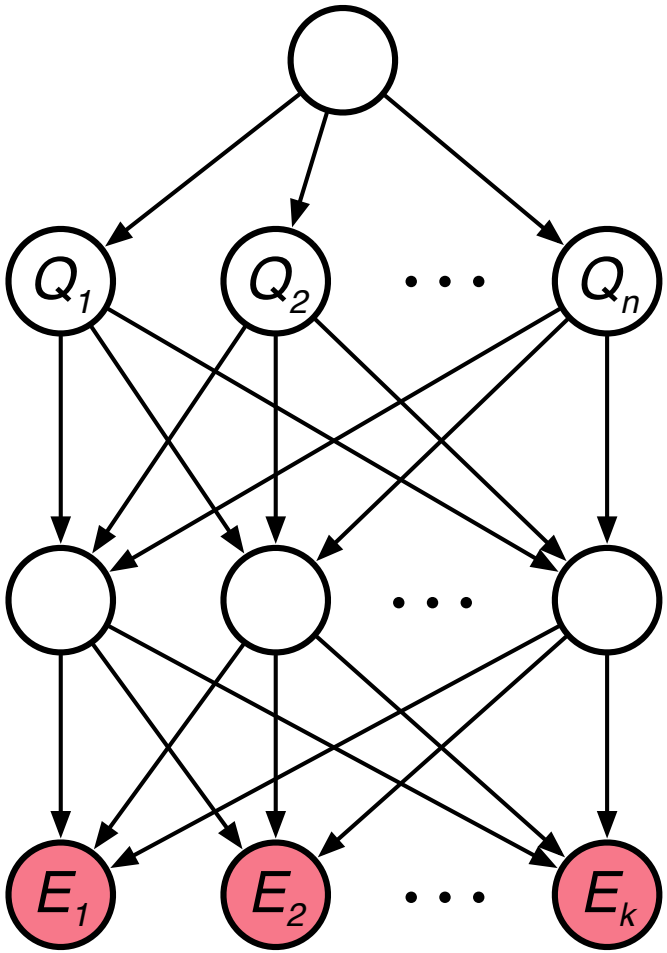

*Left* — rare evidence affects how query nodes are sampled.

*Right* — rare evidence is unlikely to occur with high probability.

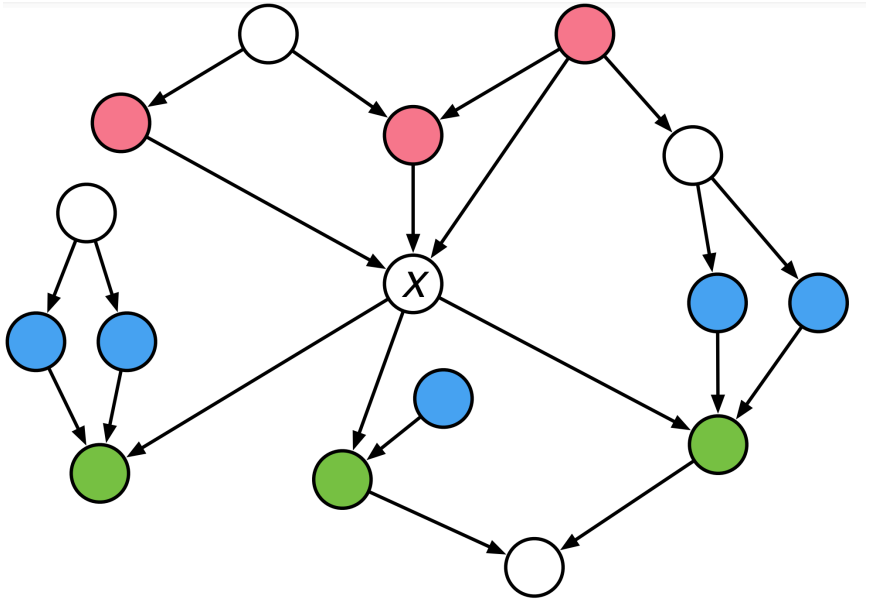
# What next?

To handle this case, especially with rare evidence, we need the evidence nodes to affect how other nodes are sampled.

We need a way to sample nodes **in any order**—not only in a forward pass when they are conditioned on their parents.



# Markov blanket




- **Definition**

  The Markov blanket $B_X$ of a node $X$ consists of its parents, children, and spouses (i.e., parents of children).

- **Theorem**

  The node $X$ is conditionally independent of **the nodes outside** its Markov blanket given the nodes inside its Markov blanket.

# Test your understanding

Let $X$ be a node in a belief network.
Let $B_X$ denote its Markov blanket (i.e., parents, children, spouses). Let $Y$ be any node such that $Y \notin X \cup B_X$.

Q. Which of these is TRUE?

A. The parents, children, and spouses of $X$ are non-overlapping sets of nodes.

B. The parents, children, and spouses of $X$ are non-overlapping in a polytree.

C. $P(X|B_X, Y) = P(X|B_X)$ is **only** guaranteed to be true in a polytree.

D. All are true.

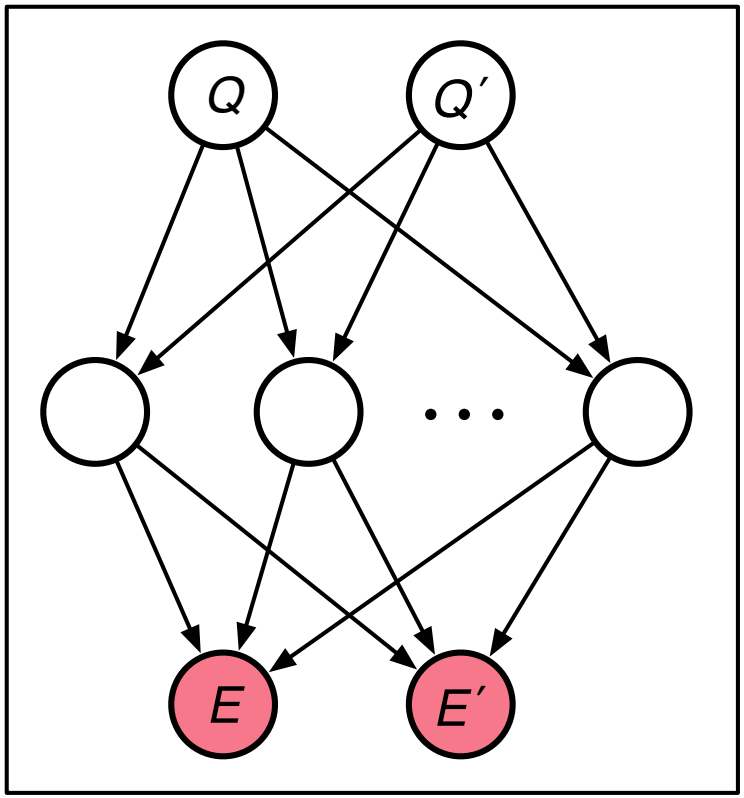
E. None are true.

# Markov chain Monte Carlo

# Approximate inference

Query nodes $Q, Q'$

Evidence nodes $E, E'$

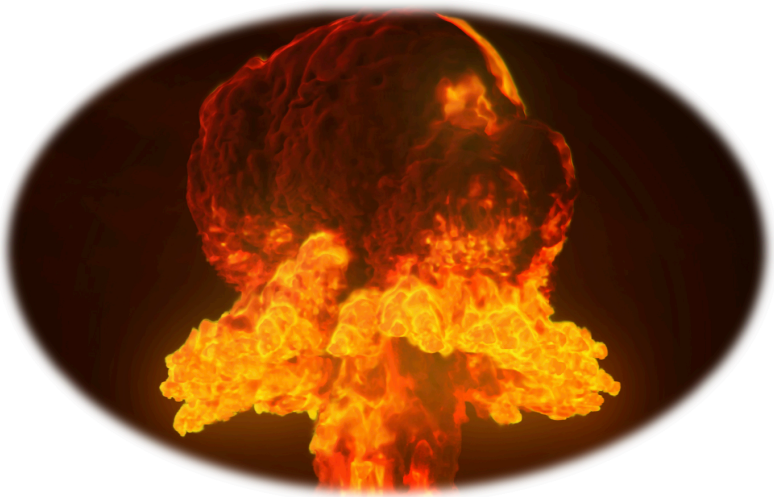



How to estimate $P(Q = q, Q' = q' | E = e, E' = e')$?

# Fun Fact!

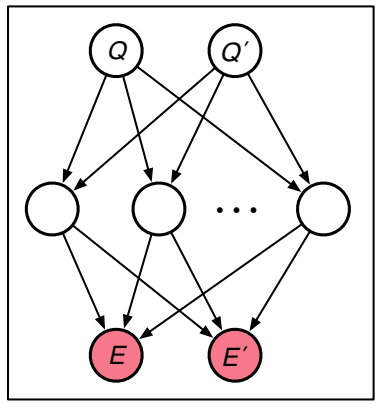
Monte Carlo methods are usually traced to physicists at Los Alamos in 1940s!

- Stanisław Ulam (inspired by solitaire!) and Von Neumann (rejection sampling).
- Interested in modeling the probabilistic behavior of collections of atomic particles.
- The term ‘Monte-Carlo’ was coined at Los Alamos.



# MCMC - Gibbs Sampling

- **Initialization**
  Fix evidence nodes to observed values $e, e'$.
  Initialize non-evidence nodes to random values.




- **Repeat $N$ times**
  Pick a non-evidence node $X$ at random.
  Use **Bayes rule** to compute $P(X|B_X)$.
  Resample $x \sim P(X|B_X)$.
  Take a snapshot of all the nodes in the BN.

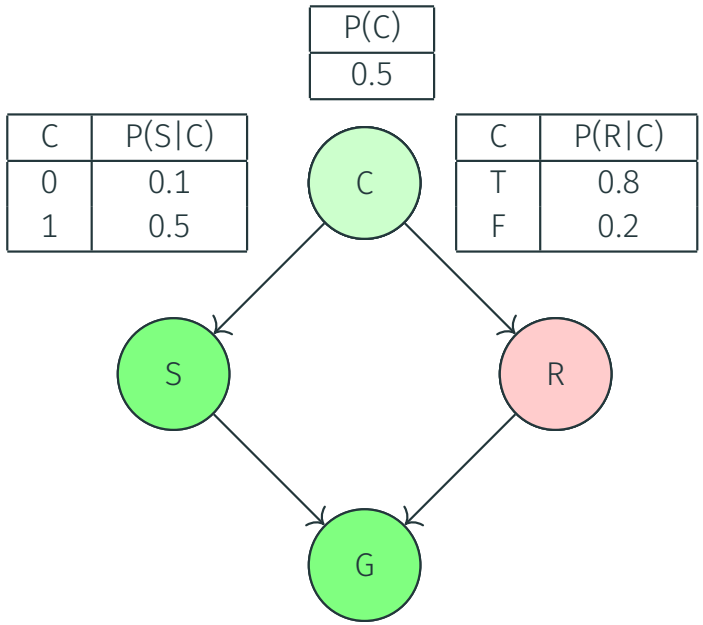
- **Estimate**
  Count the snapshots $N(q, q') \leq N$ with $Q = q$ and $Q' = q'$.

$$P(Q = q, Q' = q' | E = e, E' = e') \approx \frac{N(q, q')}{N}$$

# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

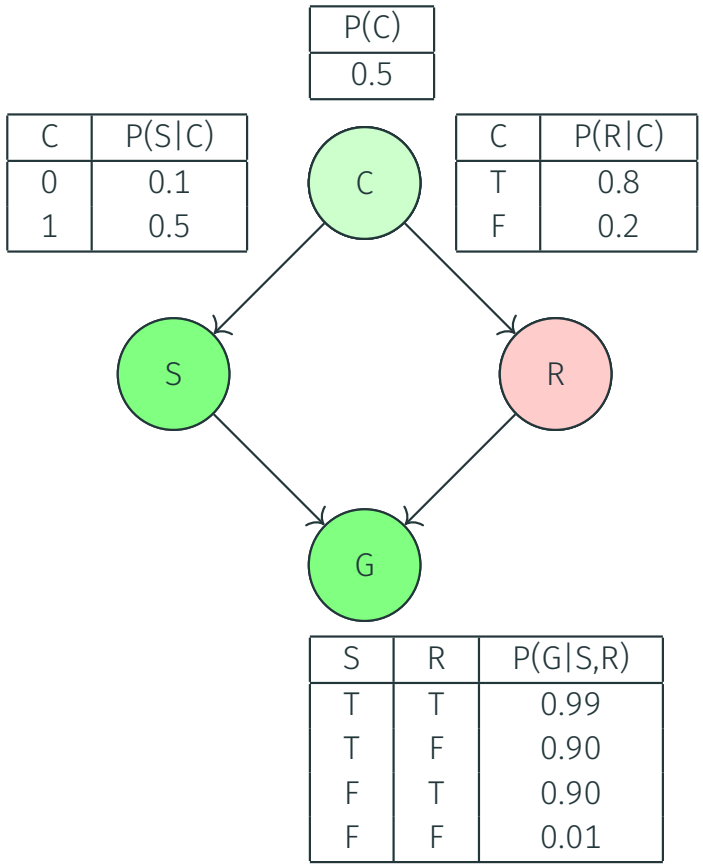
- **Initialization**
  - Set evidence: $s_1$, $g_1$
  - Randomly set non-evidence variables: $c_1$, $r_1$
- Repeat N times:
  -



# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

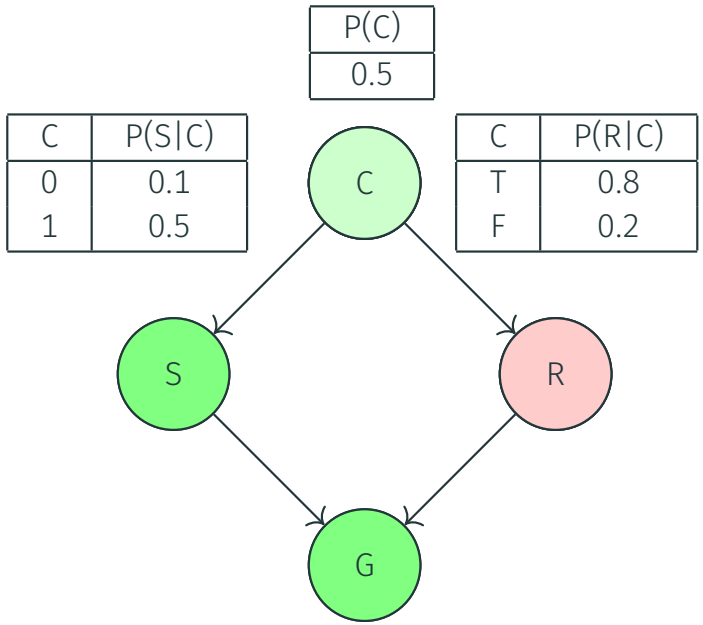
- **Initialization**
  - Set evidence: $s_1$, $g_1$
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- Repeat N times:
  - Pick variable to update from $\{R, C\}$ uniformly at random: **R**



# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

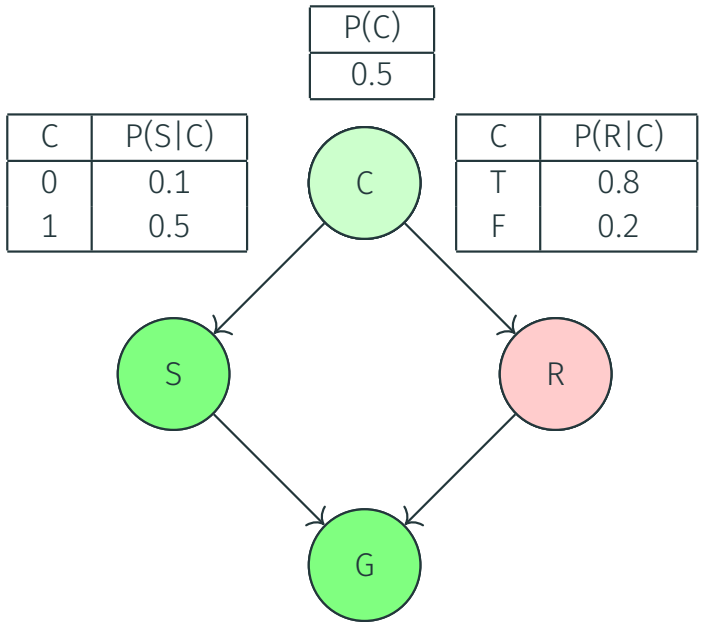
- **Initialization**
  - Set evidence: $s_1$, $g_1$
  - Randomly set non-evidence variables: $c_1$, $r_1$
- Repeat N times:
  - Compute $P(R|c_1, s_1, g_1)$ using **Bayes rule**



# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

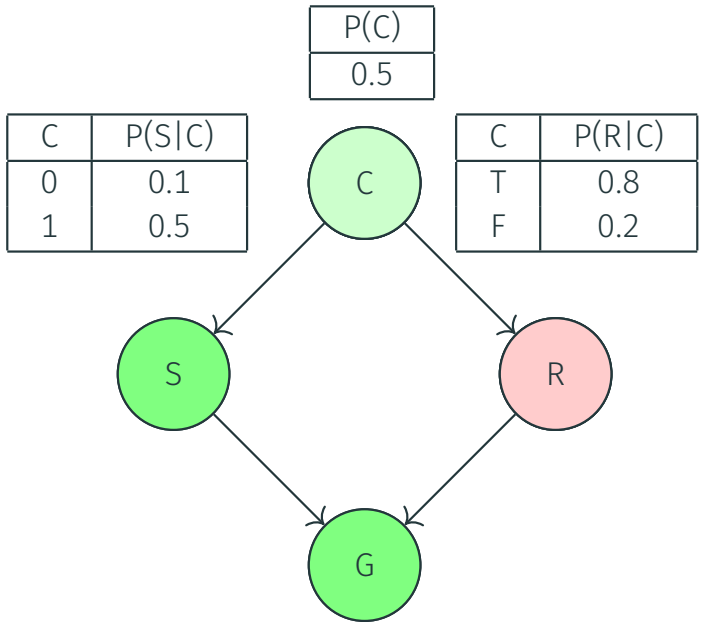
- **Initialization**
  - Set evidence: $s_1$, $g_1$
  - Randomly set non-evidence variables: $c_1$, $r_1$
- Repeat N times:
  - Sample R from $P(R|c_1, s_1, g_1)$: $r_0$
    Take a snapshot



# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

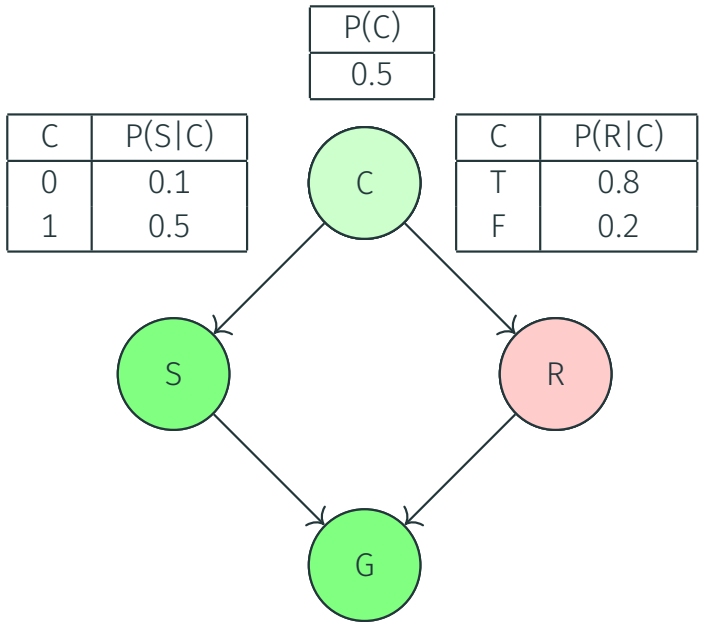
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# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

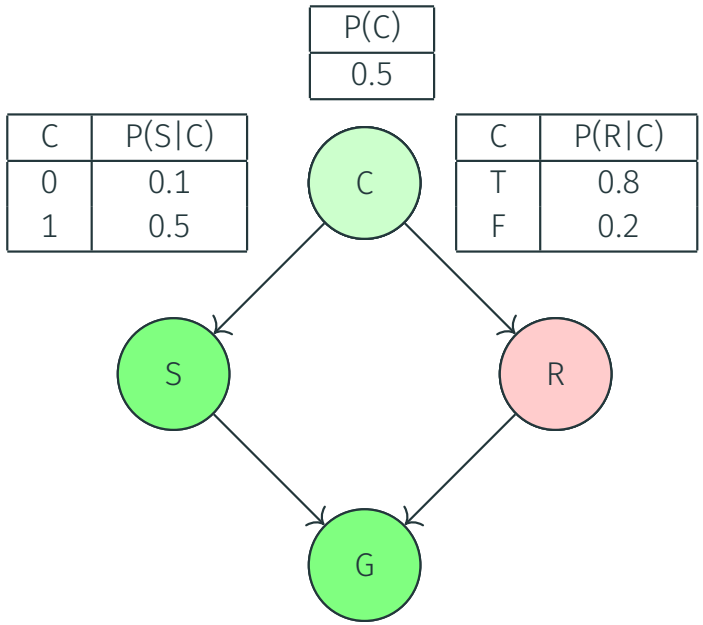
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- Repeat N times:
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# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

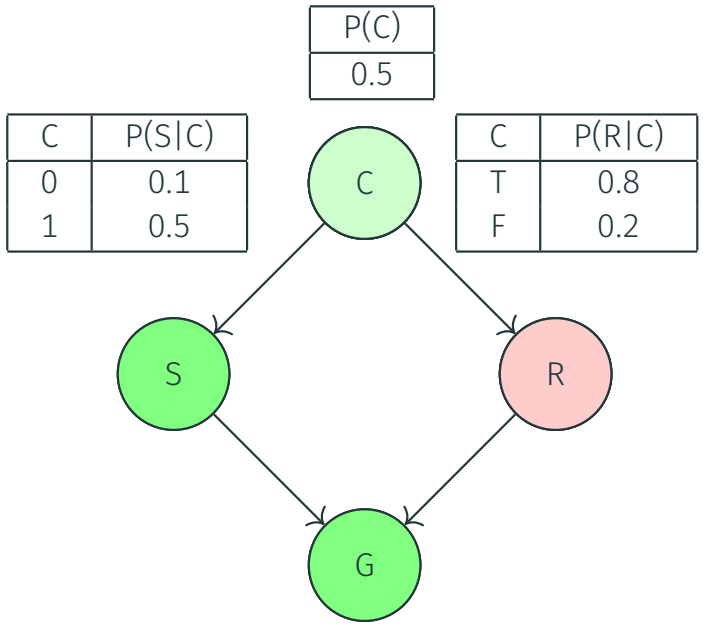
- **Initialization**
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  - Randomly set non-evidence variables: $c_1$, $r_1$
- Repeat N times:
  - Sample C from $P(C|r_0, s_1)$: $c_0$
    Take a snapshot



# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

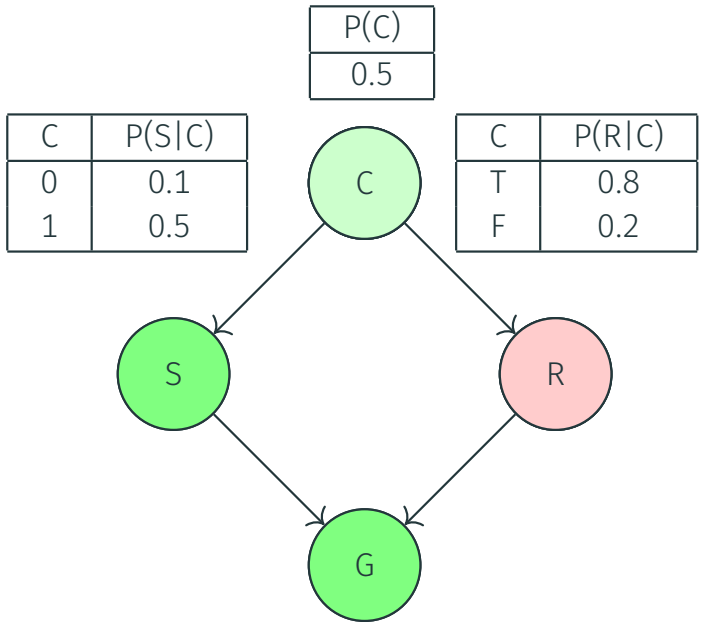
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# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

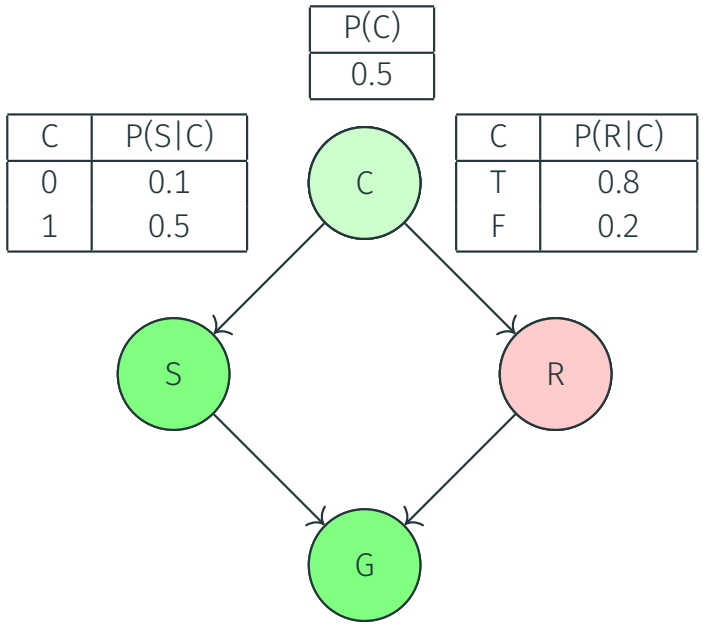
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# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

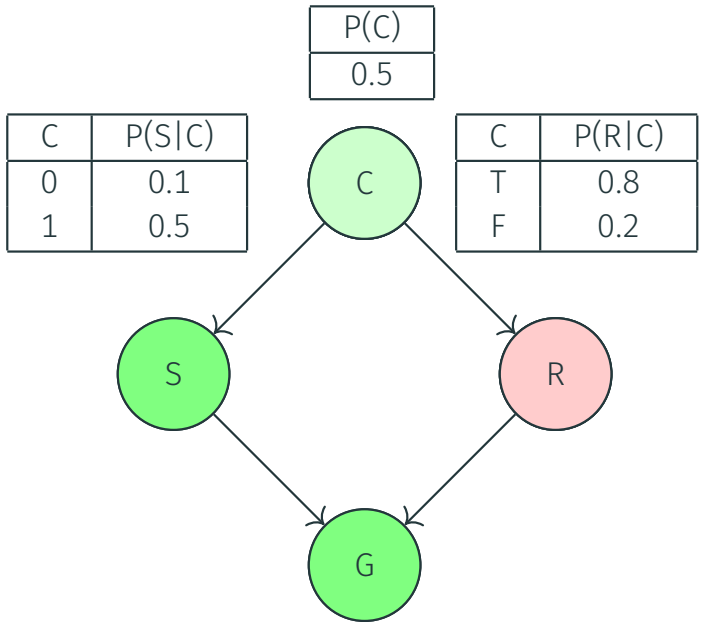
- **Initialization**
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  - Sample C from $P(C|r_0, s_1)$: $c_1$
    Take a snapshot



# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

- **Initialization**
  - Set evidence: $s_1$, $g_1$
  - Randomly set non-evidence variables: $c_1$, $r_1$
- Repeat N times:
  -
- Count the snapshots with $r_1$: $N_{r_1}$

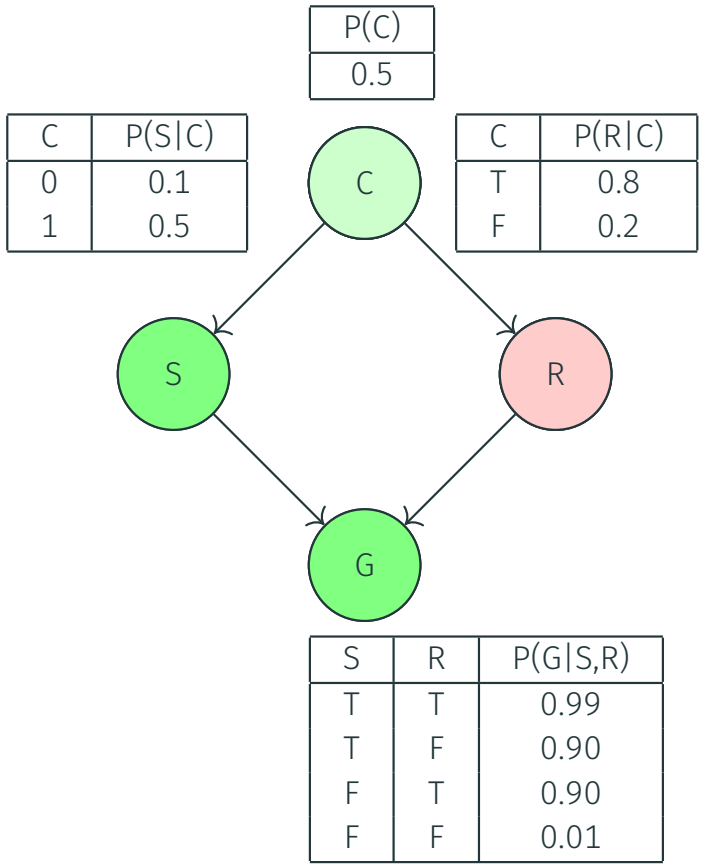


# Gibbs Sampling Example

Estimate $P(R = 1 \mid S = 1, G = 1)$

- **Initialization**
  - Set evidence: $s_1$, $g_1$
  - Randomly set non-evidence variables: $c_1$, $r_1$
- Repeat N times:
  -
- Count the snapshots with $r_1$: $N_{r_1}$

$$P(R = 1|S = 1, G = 1) \approx \frac{N_{r_1}}{N}$$



# Gibbs Sampling

**Q. (A) True or (B) False**

Gibbs MCMC could get stuck if the relationship between two random variables is *deterministic*.

# Properties of MCMC

**Under reasonable conditions...**

1. This sampling procedure defines an ergodic (**irreducibile** and **aperiodic**) Markov chain over the non-evidence nodes of the BN.
2. The stationary distribution of this Markov chain is equal to the BN's posterior distribution over its non-evidence nodes.
3. Theoretical guarantees for mixing time, in practice we use burn in time.
4. The estimates from MCMC converge in the limit:

$$\lim_{N \to \infty} \frac{N(q, q')}{N} \rightarrow P(Q = q, Q' = q' | E = e, E' = e')$$

# MCMC versus likelihood weighting (LW)

- How they sample

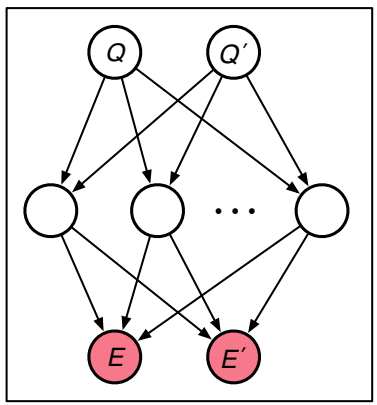
$$\left.\begin{matrix}\text{LW}\\ \text{MCMC}\end{matrix}\right\} \text{ samples non-evidence nodes from } \begin{cases} P(X|\mathrm{pa}(X)) \\ P(X|B_X) \end{cases}$$

- Cost per sample

  LW can read off $P(X|\mathrm{pa}(X))$ from each CPT.
  MCMC must compute $P(X|B_X)$ before each sample.

- Convergence

  LW is slow for rare evidence in leaf nodes.
  MCMC can be much faster in this situation.



# That’s all folks!